"Slow Is Smooth, Smooth Is Fast": Artificial Intelligence-Driven Suturing Simulation for Cardiothoracic Surgical Training.

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Abstract

To develop and evaluate a suturing simulator using Artificial intelligence (AI)-driven computer vision for objective, high-resolution assessment of technical performance in cardiothoracic surgery, applied to a novel 3D-printed simulator across different levels of surgical expertise. Prospective study using a novel 3D-printed suturing simulator with targets positioned at multiple depths and angles relevant to cardiothoracic surgery. Participants performed standardized running baseball sutures on 2 planes while procedural videos and post-completion images were recorded. Supervised machine learning-based video tracking (MediaPipe) and classical image processing were used to derive temporal metrics, hand path length, acceleration variability, movement irregularity, and suture uniformity. These metrics were then compared between attending surgeons and trainees. Brigham and Women's Hospital, an academic tertiary care center in Boston, Massachusetts. Ten attending cardiac surgeons and 10 surgical trainees. AI-assisted video analysis differentiated experts from trainees with high precision. Attending cardiac surgeons performed significantly faster than trainees in needle loading (1.9 seconds vs. 2.5 seconds, $p < 0.01$) and stitch execution (4.6 seconds vs. 6.4 seconds, $p < 0.01$), resulting in shorter ring completion times (252 seconds vs. 312 seconds, $p < 0.02$). Attending surgeons also demonstrated superior motion efficiency and less irregular movements during key maneuvers, characterized by reduced overall path length (6,714 vs. 10,154 mm, $p = 0.01$), lower average acceleration (20 vs. 25 mm/s2, $p = 0.01$) and reduced vibrational motion patterns (26,786,414 vs. 56,008,703 mm2/s4, $p = 0.01$). Image analysis confirmed that attending surgeons produced significantly more uniform sutures with lower variability in inter-stitch distances (standard deviation [SD] 1.72 vs. 2.14 mm, $p = 0.02$; uniformity 68.9% vs. 63.1%, $p = 0.03$). AI-based assessment on an anatomically relevant simulator distinguished experts from novices using granular and mechanistic metrics of suturing skill across AI-segmented motion phases. Experts demonstrated more deliberate, controlled, and smoother movements while completing tasks in less time.

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